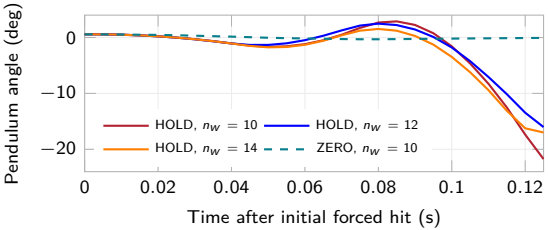
Furuta: the three published HOLD attacks

$\ell = 25,\ \hat{p}_m = 0.25,\ \bar{a} = 4,\ \alpha = 0.01,\ T_s = 5\,\mathrm{ms}.$ The same schedule and initial state drive both policies.





Supplied dashboard: schematic angle view.

All pass watchdog and binomial. Geometric tests reject all three: $\chi^2 = 187.30,\ 86.22,\ 94.78 > 13.277$.

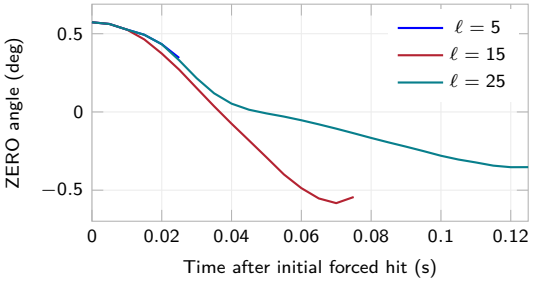
Window	Misses	ω_{HOLD} (rad^2)	ω_{ZERO}, same attack	Peak $\lvert\theta\rvert$, HOLD
10	15	0.144120	-7.28×10^{-6}	21.75°
12	14	0.078222	-5.72×10^{-6}	16.03°
14	14	0.087986	-7.36×10^{-6}	17.00°

Validation: same-schedule replay supports the paper's actuation-policy sensitivity.

Paper Table 1 / Fig. 4; supplied ZIP: dashboard, Python replay and validated statistics.

Furuta: horizon and detector validation

The published ZERO cases use the combined detector with $n_w = 20$. The earlier 23-period replay uses binomial detection with $n_w = 10$.



Case	ω_{HOLD}	ω_{ZERO}
Paper, $\ell = 5$	-8.80×10^{-6}	2.74×10^{-5}
Paper, $\ell = 15$	1.18×10^{-4}	8.46×10^{-5}
Paper, $\ell = 25$	9.01×10^{-6}	2.88×10^{-5}
Earlier, $\ell = 23$	4.24×10^{-2}	-4.93×10^{-6}

For the three published cases, $\chi^2 = 6.33,\ 12.83,\ 12.11$: all pass the combined constraints.

The earlier replay has $\chi^2 = 126.11$. It passes binomial detection and fails the combined detector.

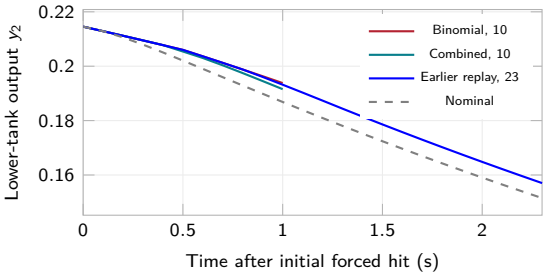
Validation: these cases show that impact depends on the schedule and active detector. A longer horizon does not imply larger terminal impact.

Separate attack problems; values use rad^2. The earlier 23-period replay has no optimality certificate.

Paper Sec. 5.4.1 / Fig. 2; earlier reviewed Furuta presets; exact REP v1.2 matrices.

Quadruple tank: detector comparison and replay

Paper HOLD model, $T_s = 0.1\,\mathrm{s}$, $\hat{p}_m = .25$, $\bar{a} = 4$, $\alpha = .01$. Two 10-period searches enumerate all 2^{10} schedules.



Case	HOLD ω	χ^2
Binomial, $\ell = 10$	0.003162	338.78
Combined, $\ell = 10$	0.002074	10.71
Earlier, $\ell = 23$	0.001549	15.00

All pass their watchdog and binomial constraints. Only the combined-search case passes $\chi^2 \leq 13.277$.

Adding the geometric constraint reduces the 10-period maximum by **34.4%** for this configuration.

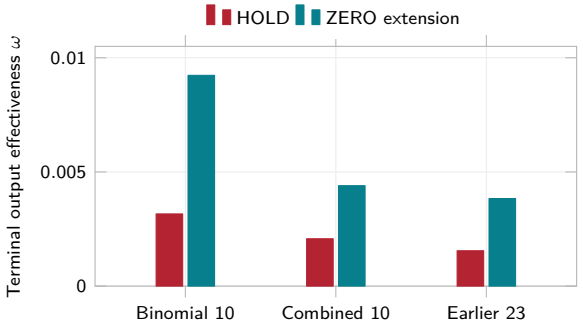
Validation: this finite-horizon result supports the paper's detector comparison in Fig. 3.

The curves show output deviations from equilibrium, not absolute water heights. The earlier 23-period schedule has no optimality certificate.

Paper Figs. 2–3; earlier reviewed tank presets, independently replayed with REP v1.2.

Quadruple tank: the ZERO extension changes the result

Each pair uses the same schedule and initial state. HOLD is the paper model. Tank ZERO is a project extension.



Exact REP tank HOLD matrices; ZERO miss-input block extension; all three earlier tank cases.

All three tank schedules produce more terminal damage under ZERO. The increases are $2.92\times$, $2.12\times$, $2.48\times$, respectively.

Tank ZERO sets the next input *deviation* to zero. It does not mean switching the physical pumps off.

The paper's tank output is

$$y_k = \tfrac{1}{2}[x_{1,k}, x_{2,k}]^\top, \qquad \omega_y = \tfrac{1}{4}\omega_{\mathrm{REP}}.$$

Interpretation: the better actuation policy depends on the plant and tested attack. The Furuta comparison does not establish a universal ZERO rule.

Defender extension: choosing a configuration

The paper optimizes attacks for a fixed configuration. Our outer sweep compares defender choices using the original optimizers.

$$W(d,p) = \max_{\ell \in \mathcal{L}} \omega^*(d,p,\ell), \qquad d^* = \arg\min_{d} \max_{p \in \mathcal{P}} W(d,p).$$

Quantity	Supplied ZIP settings
Policy	Furuta HOLD / ZERO
Window n_w	$10, 12, 14, 16, 18, 20$
Significance α	$0.005, 0.01, 0.05$
Nominal scenarios $\mathcal{P}$	$0.09, 0.17, 0.25$
Consecutive-miss cap	$\bar{a} = 4$
Detector	Binomial by default
Quick horizons $\mathcal{L}$	$5, 10, 15, 20, 25$
Full horizons	$1, \ldots, 25$

Scope: 36 defender choices in each nominal scenario, or 540 optimization runs on the quick grid. The full grid needs 2,700 runs.

$\hat{p}_m$ represents measured nominal behavior. It is a scenario parameter, not a free protection knob.

Rank configurations only after checking all requested horizons and scenarios, solver status, and objective replay.

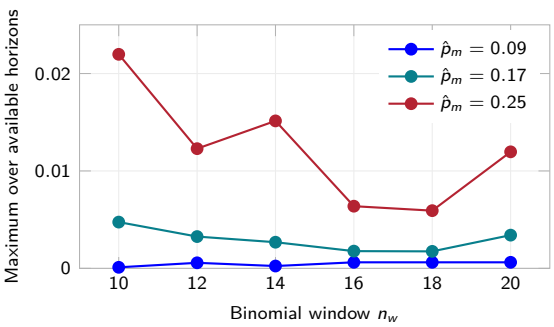
Empirical false alarms must accompany damage comparisons. α alone is not the false-alarm rate over a whole session.

Contribution: select the policy, window and significance level against the tested attack family. A finite sweep does not certify physical safety.

Supplied ZIP notebook Sec. 7; rep_defender_driver.py. The ZIP contains no completed sweep results.

Defender validation: horizon coverage changes the conclusion

The earlier linked notebook saved 432 of 540 quick-grid runs. All 108 runs at $\ell = 25$ failed, so its maxima cover $\{5, 10, 15, 20\}$.



HOLD, $n_w = 10$, $\hat{p}_m = .25$, $\alpha = .01$:

Evidence	Effectiveness ω
Saved maximum, $\ell \leq 20$	≈ 0.021973
Table 1 replay, $\ell = 25$	0.144120

The known 25-period attack is **about 6.56 times** the displayed maximum over available horizons.

Supported finding: window choice changes observed attack impact, and the trend is not monotone.

Remaining validation: finish the missing solves before claiming the full-grid optimum or a safety guarantee.

Plot: HOLD, $\alpha = .01$, $\bar{a} = 4$, binomial detector. Values reproduce the saved notebook's rounded table, not a new sweep.

Earlier linked notebook, saved Sec. 5 outputs; ZIP Table 1 replay supplies the 25-period check.